xAI-Proj-B
Bachelor Project
Chair of Explainable Machine Learning (xAI)

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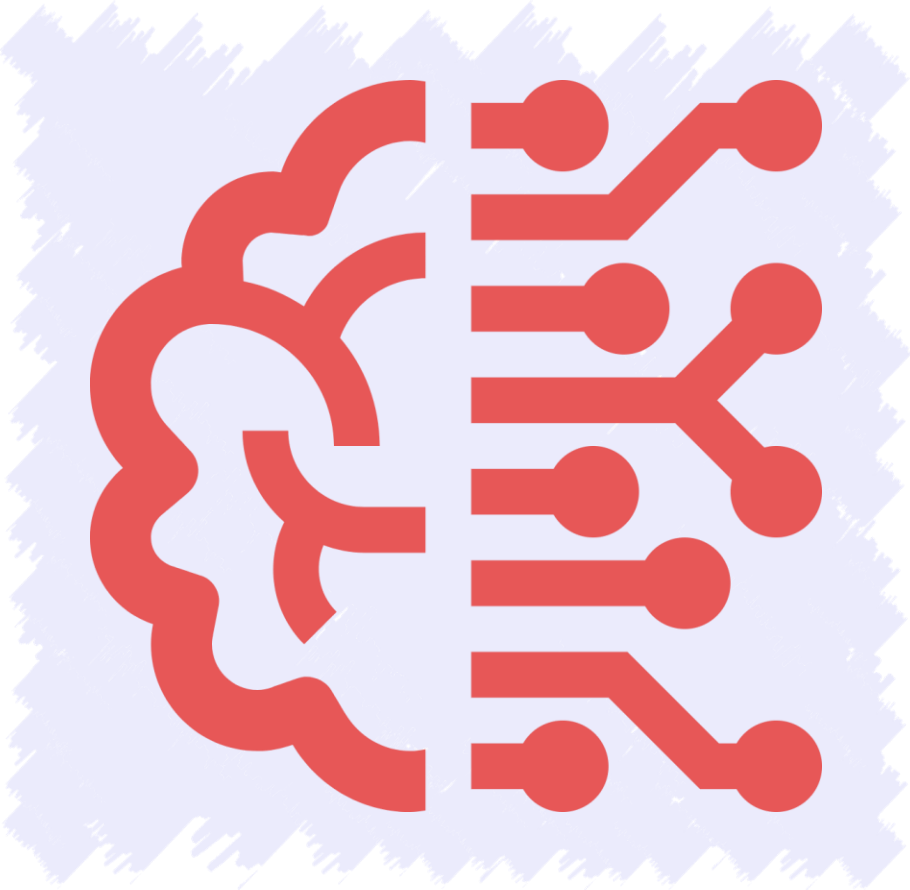
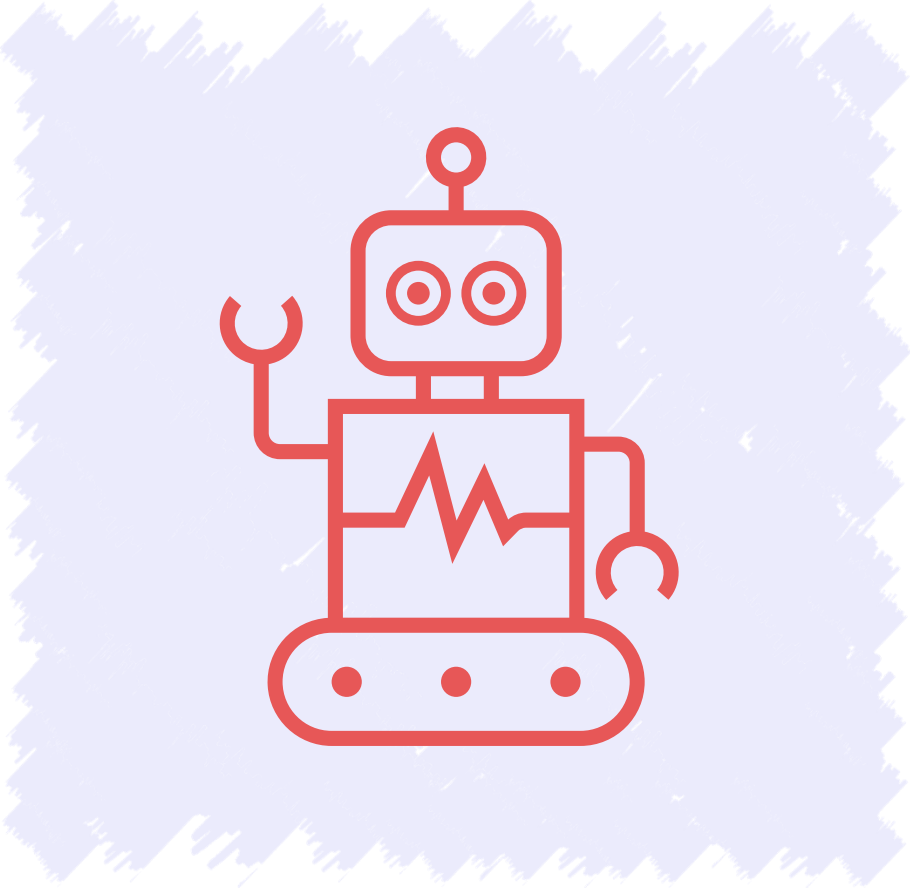
Deep Learning Introduction



What is Artificial Intelligence (AI)?



How can we define AI?

"Think like humans"

"Act like humans"

"AI is the ability of a system (or a robot controlled by a system) to imitate human behavior and human abilities such as problem solving, decision making, learning, planning, or creativity."

"Think rationally"

"Act rationally"

Ref.: [Russel_AI]
https://www.ibm.com/cloud/learn/what-is-artificial-intelligence
https://www.europarl.europa.eu/news/de/headlines/society/20200827STO85804/was-ist-kunstliche-intelligenz-und-wie-wird-sie-genutzt

So, what exactly is AI?

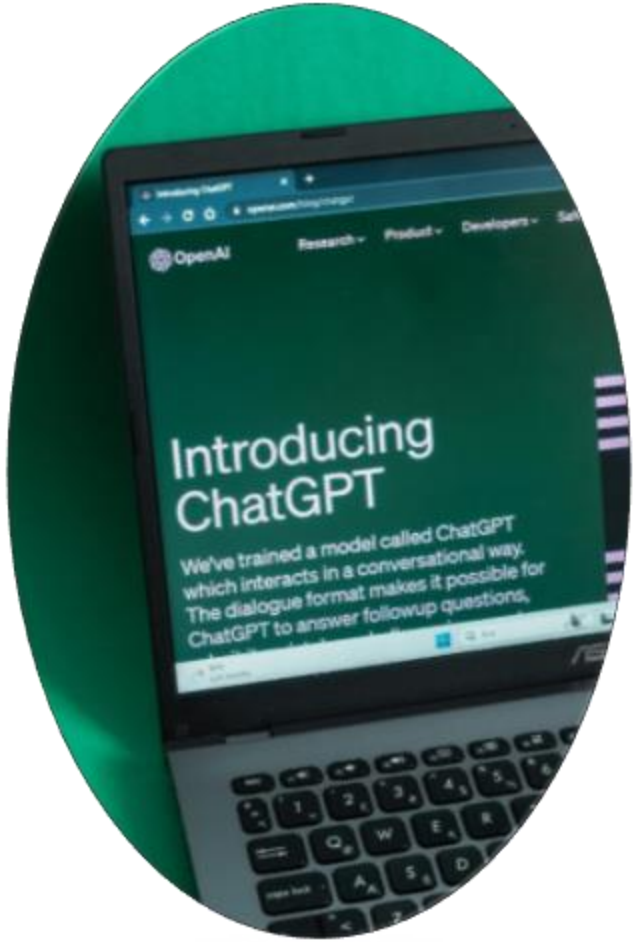


Photo by Beyzaa Yurtkuran on Pexels

by LJ on Pexels

Photo by Jonathan Borba on Pexels

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We need more definitions!

Weak or narrow AI

- Solve a specific problem
  - Self-driving cars
  - Language service systems
  - Recycling systems
  - …
- Status Quo
  - Major advances
  - Deep Learning-based
  - Large amounts of data
  - High computing power
  - Changes entire industries

Strong or general AI (AGI)

- Can adapt to new situations and tasks
- Imitation of human brain capabilities
  - Self-awareness
  - Passes the Turing test
- Status Quo
  - Large language / foundation models deliver impressive results
  - But little conceptual progress
  - There is no concensus on how general AI can be achieved
  - **It is unclear how long it will take!**

So, what do we mean when we talk about AI today?

Artificial Intelligence

Systems that exhibit "human-like" intelligence by analyzing their environment and taking actions autonomously.

„The goal"

Machine Learning

Statistical learning methods that build models from data

- Often relies on manual feature engineering
- Prediction model instead of complex rules

„ A category of learning methods for fulfilling AI"

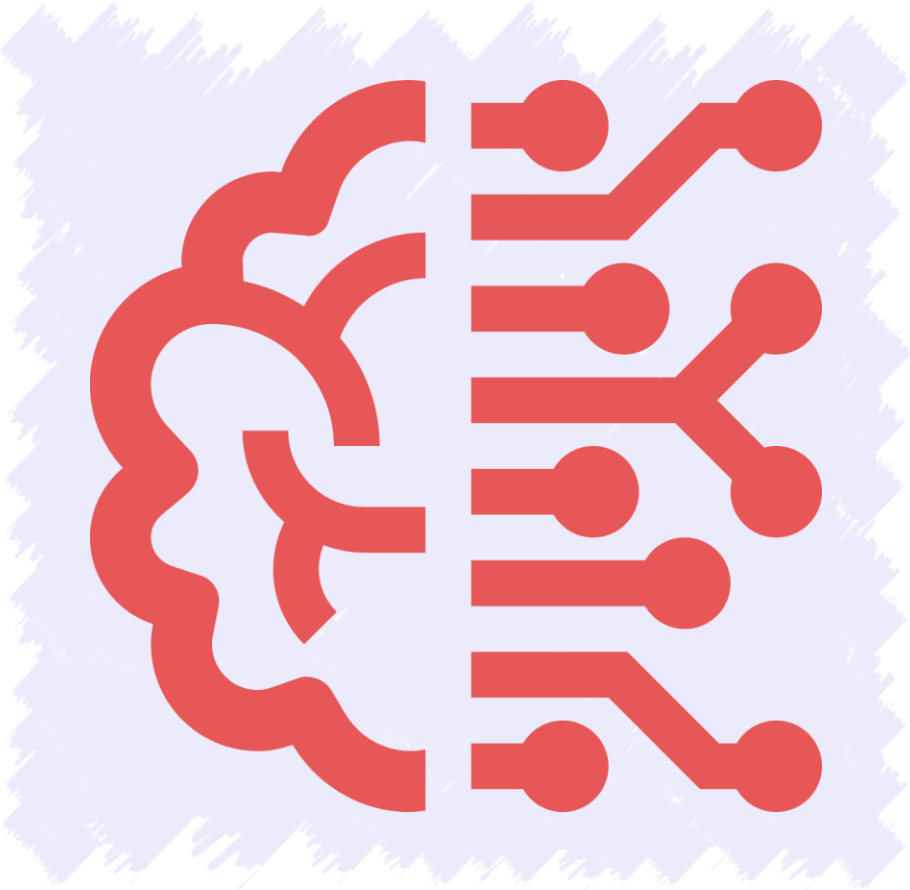
Deep Learning

Algorithms to directly learn statistics/models from data

- Inspired by the human brain
- End-to-end optimization (no feature engineering)
- Compute intensive

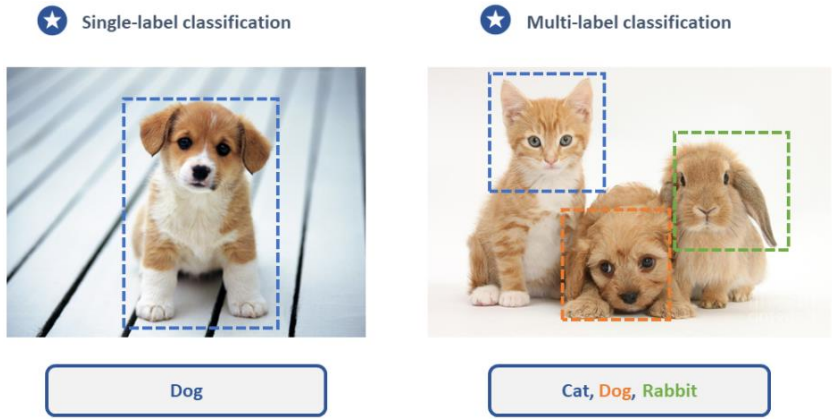
„A distinct approach in the field of machine learning"

Deep Learning



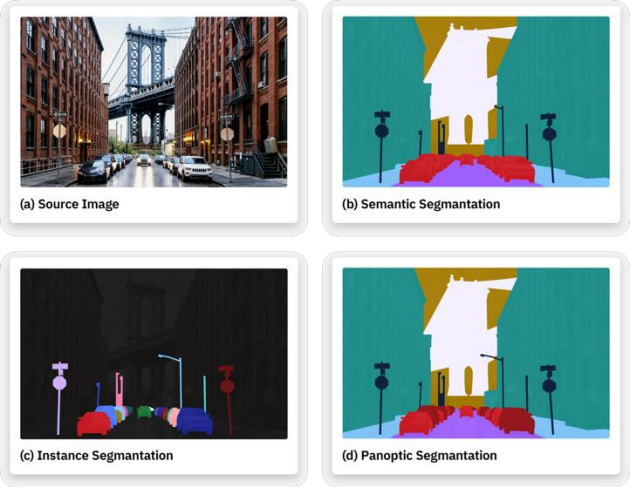
Applications

Image Classification



AI Multiple Research: Image Classification: 6 Applications & 4 Best Practices
https://research.aimultiple.com/image-classification/ [Accessed: 05/30/2025

Image Segmentation



IBM Think: What is instance segmentation?
https://www.ibm.com/think/topics/instance-segmentation?
[Accessed: 05/30/2025]

Object Detection

IBM Think: What is image recognition?
https://www.ibm.com/think/topics/image-recognition [Accessed: 05/30/2025]

Natural Language Processing

Photo by Beyzaa Yurtkuran on Pexels

Applications

Image Generation

Gemini: What Gemini can do
https://gemini.google/overview/image-generation/?hl=en [Accessed: 05/30/2025]

But how does it work?

Deep Learning explained!

Image classification

Deep Learning explained!
Image classification

Iskender, Dirik. Deep Learning einfach gemacht. Erschienen in: medium.com (2018).
Janssen, D. Machine Learning in der Schule–Ein praxisorientierte Einführung in künstliche neuronale Netze, Gesichtserkennung und Co. Science On Stage Deutschland, Berlin (2020).

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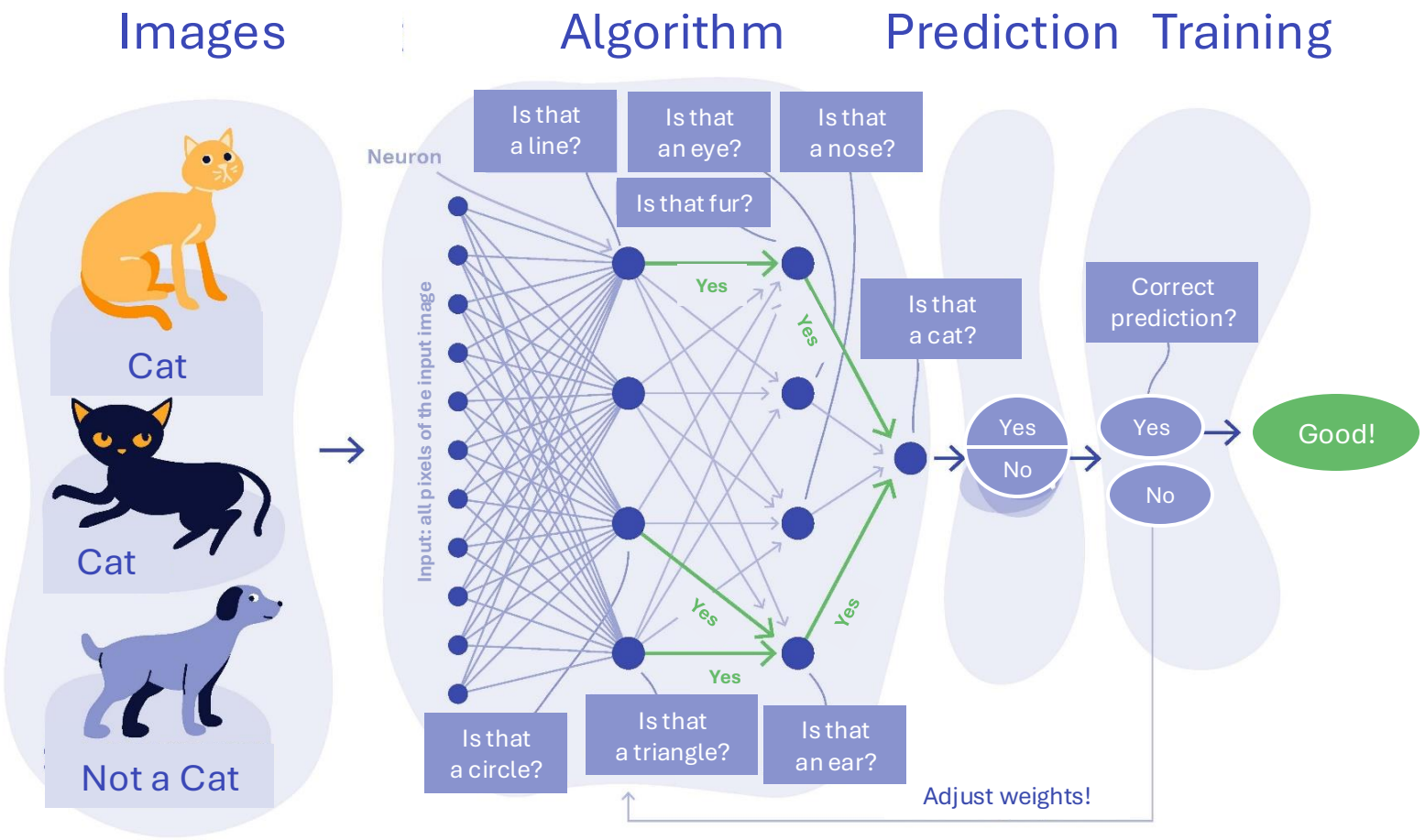
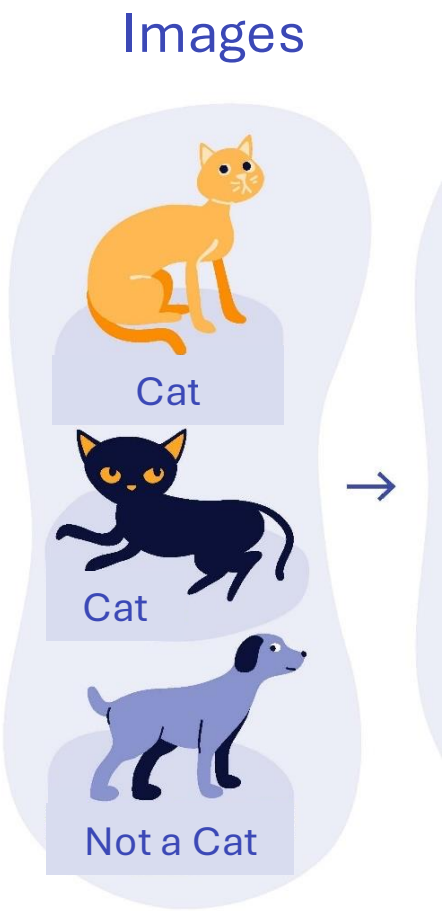
Deep Learning explained!
Image classification



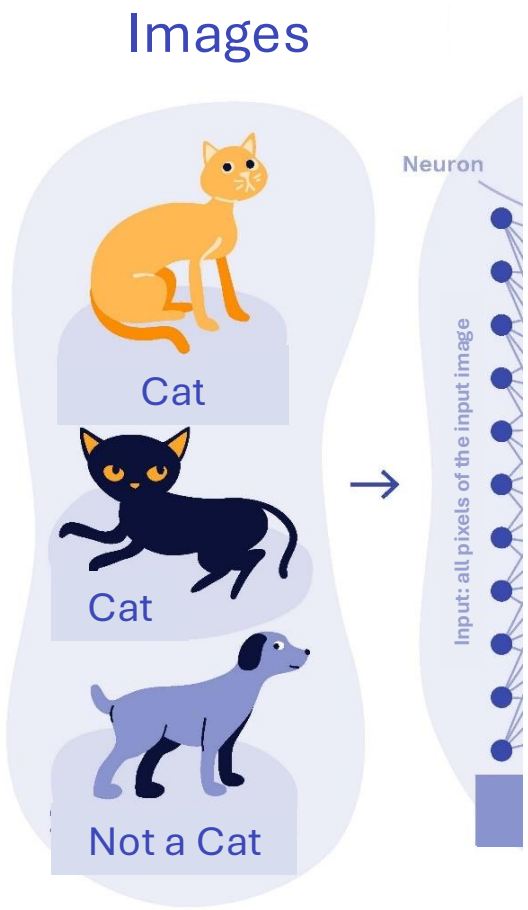
Deep Learning explained! Image classification



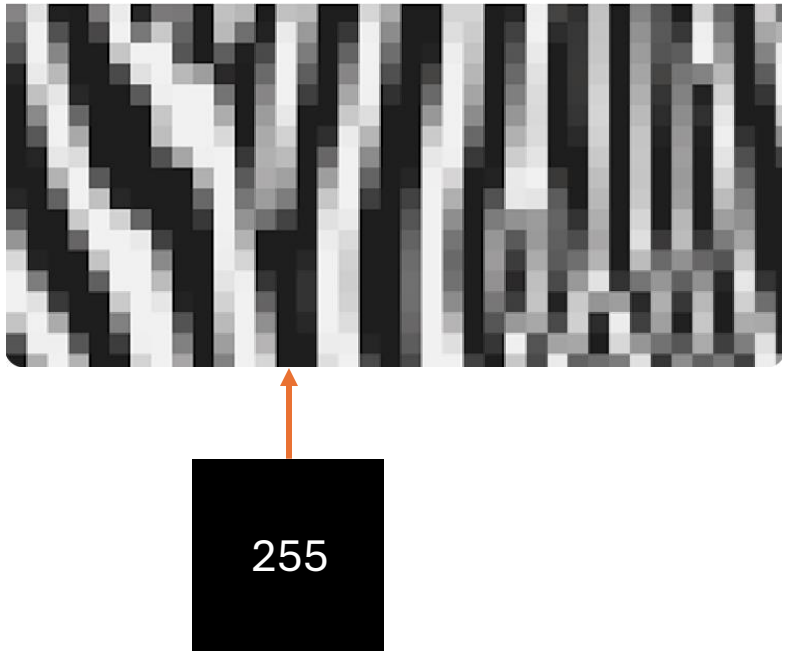
- What is supervised learning? => The images have class annotations.

- Artificial neural networks learn the connection between images and their annotations

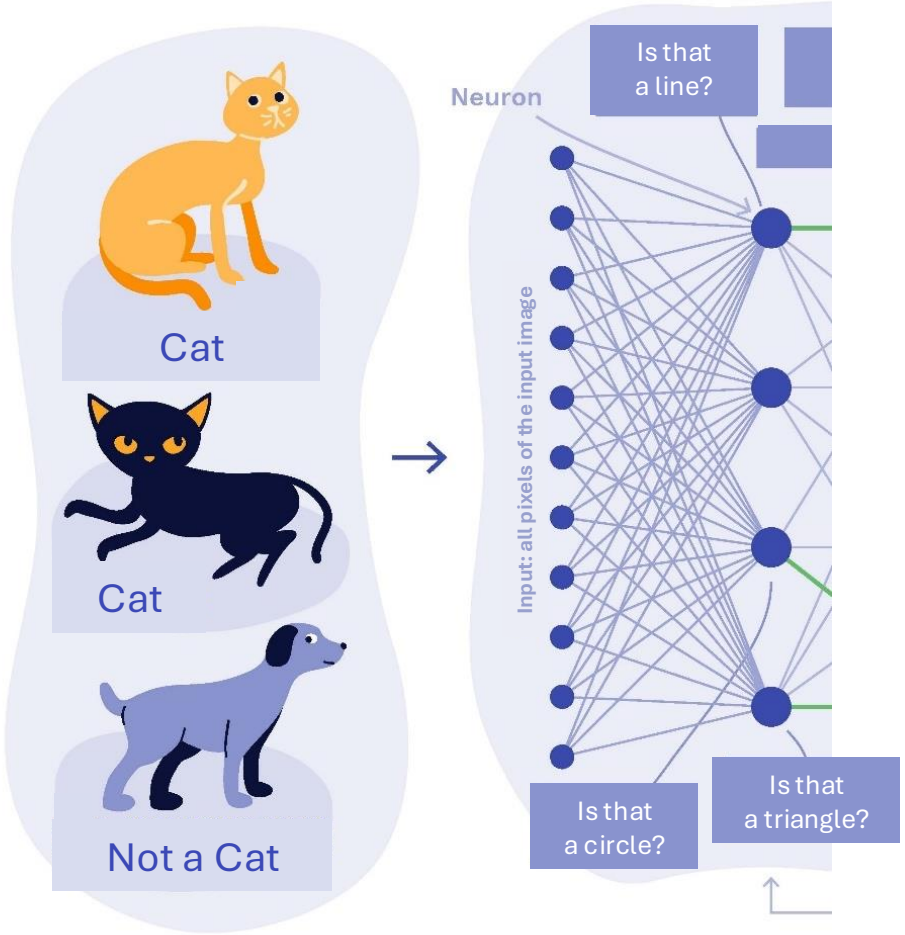
Deep Learning explained! Image classification



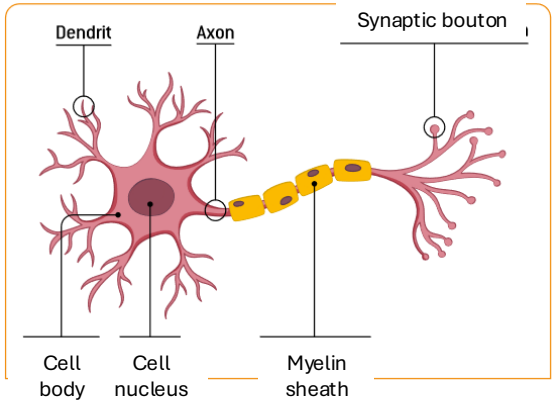
- What is the input of the neural network for image classification? => Pixel values



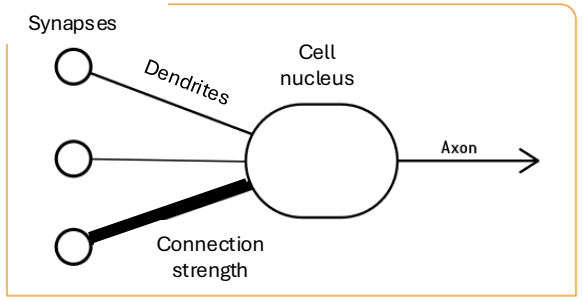
Deep Learning explained!
Image classification



- What is a Neuron?
- How are they connected?

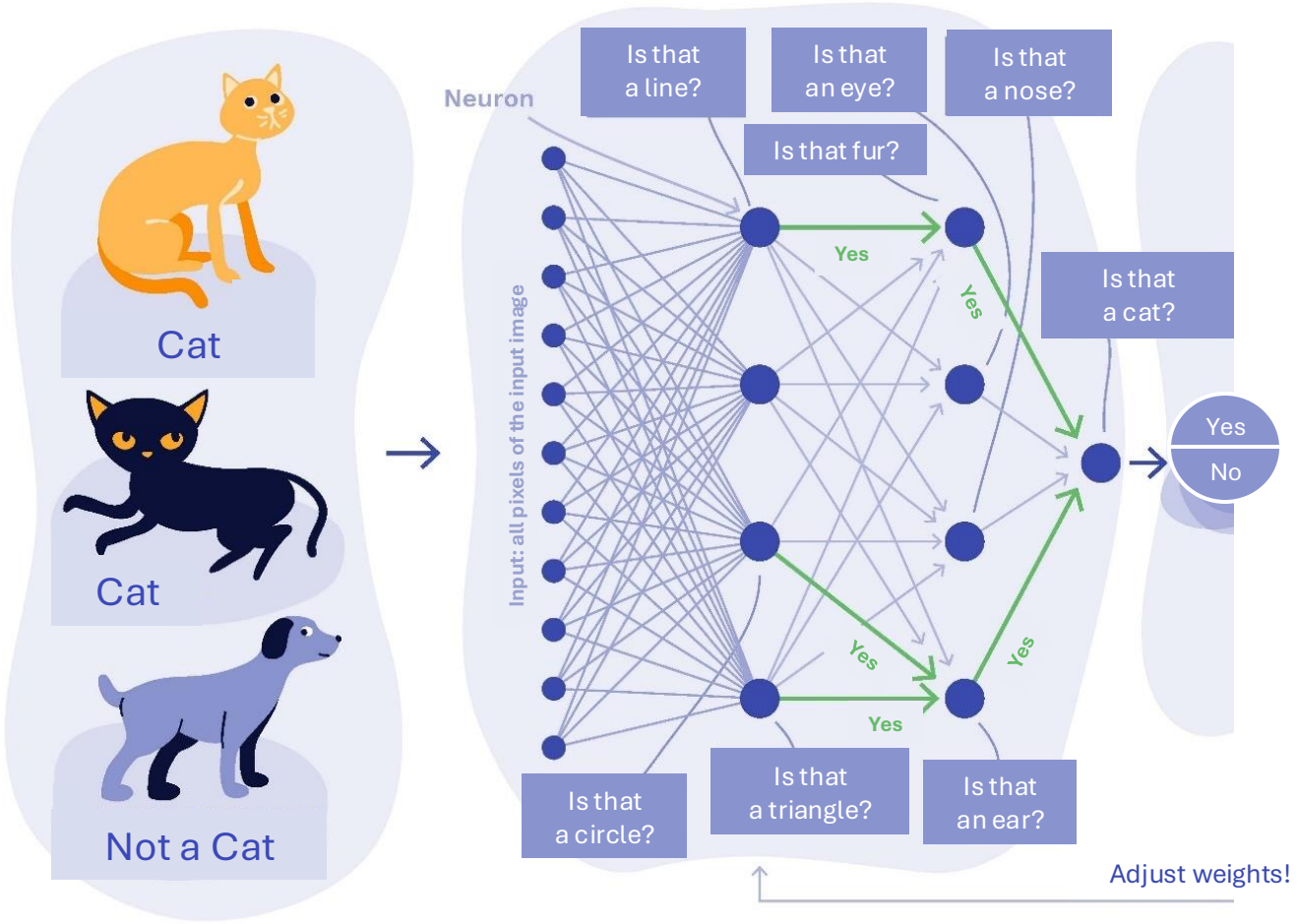


Biological Neuron



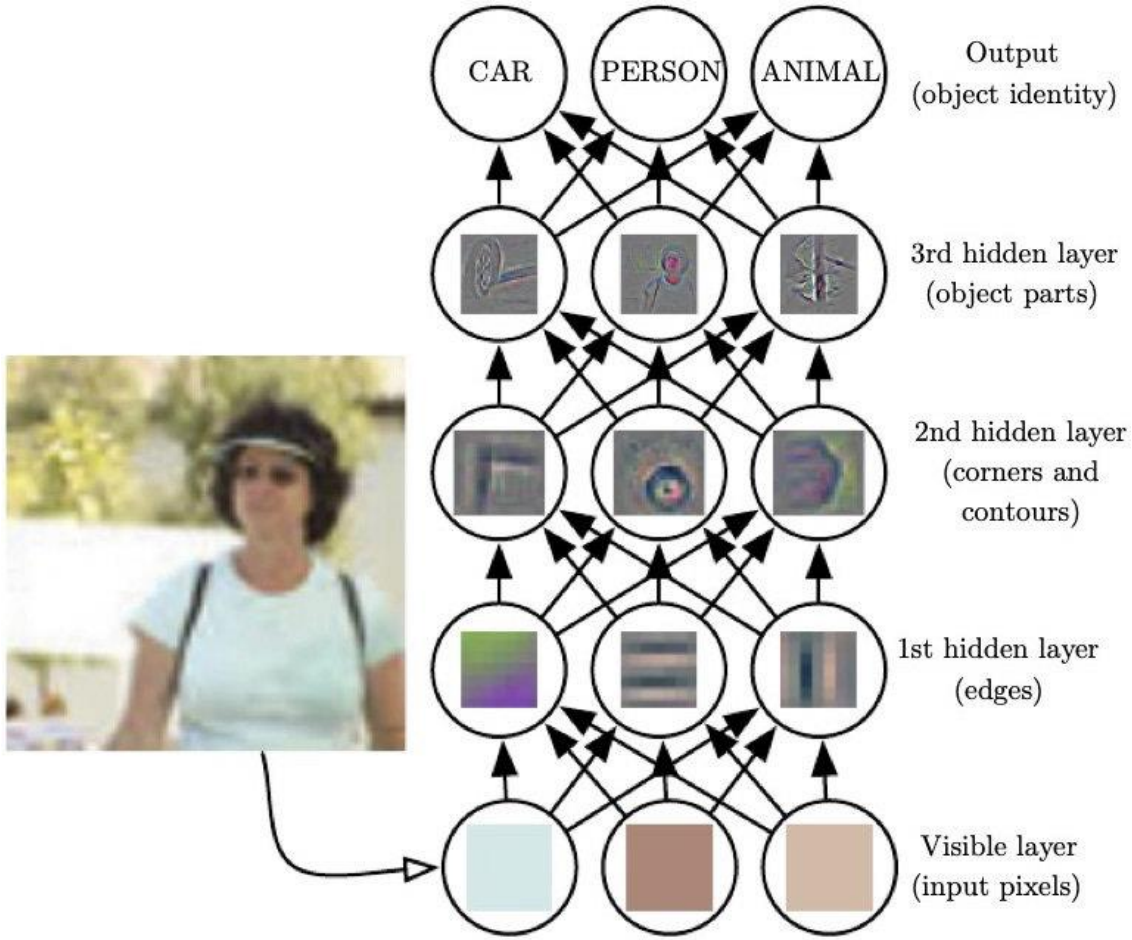
Artificial Neuron

Deep Learning explained! Image classification



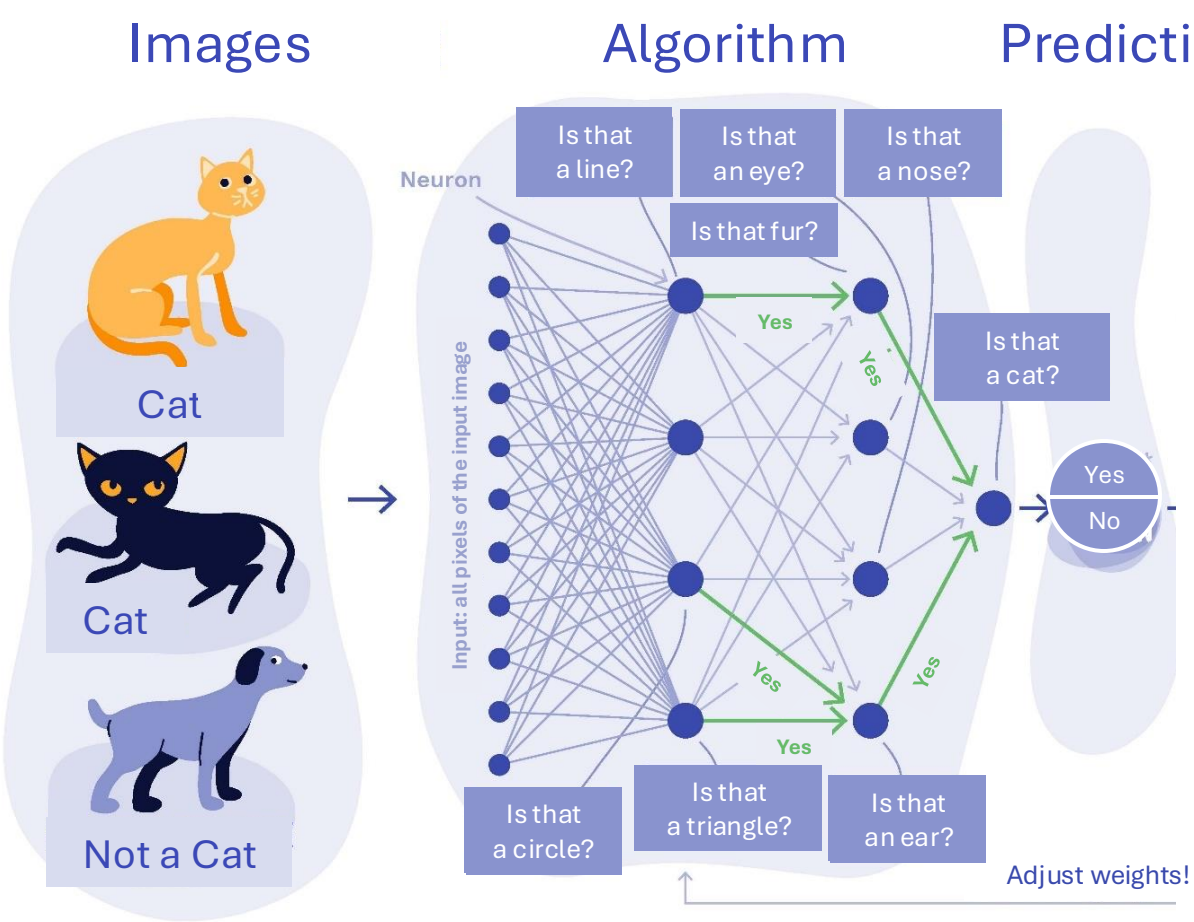
- What do the different layers in a neural network learn?
- What happens when a network is deeper?

Deep Learning explained!
Image classification



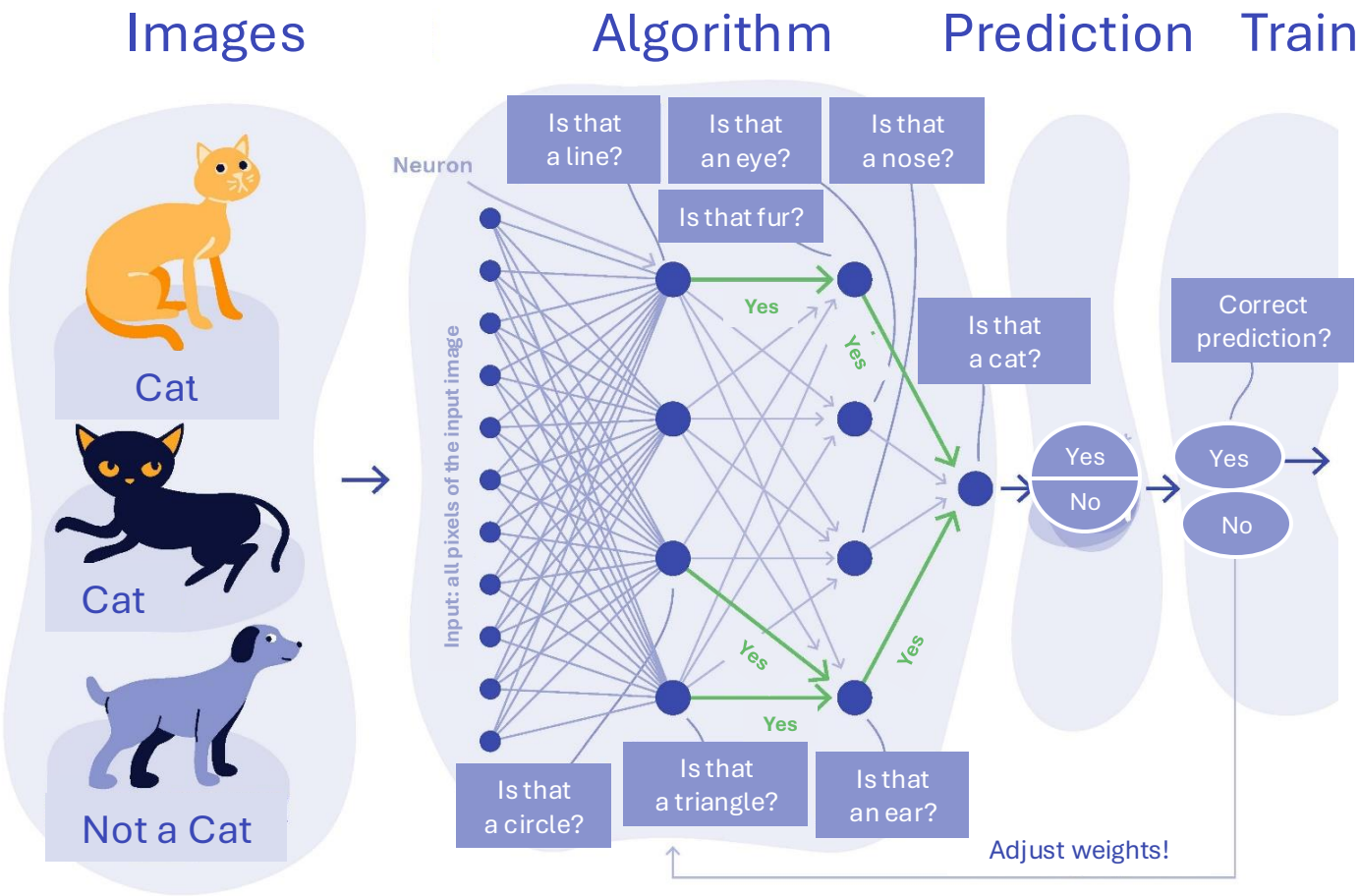
Lee, Rosemary. (2020). Machine Learning and Notions of the Image.

Deep Learning explained!
Image classification



- Classification

Deep Learning explained!
Image classification

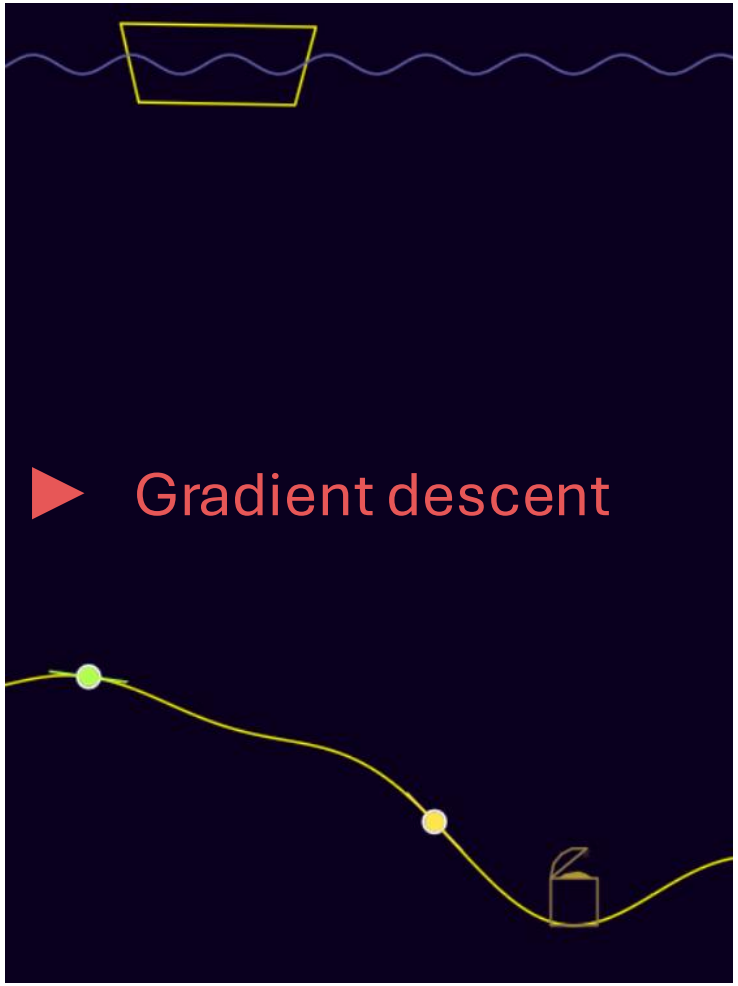


After the classifikation:

- 1. Error calculation
- 2. Adjustment of weights

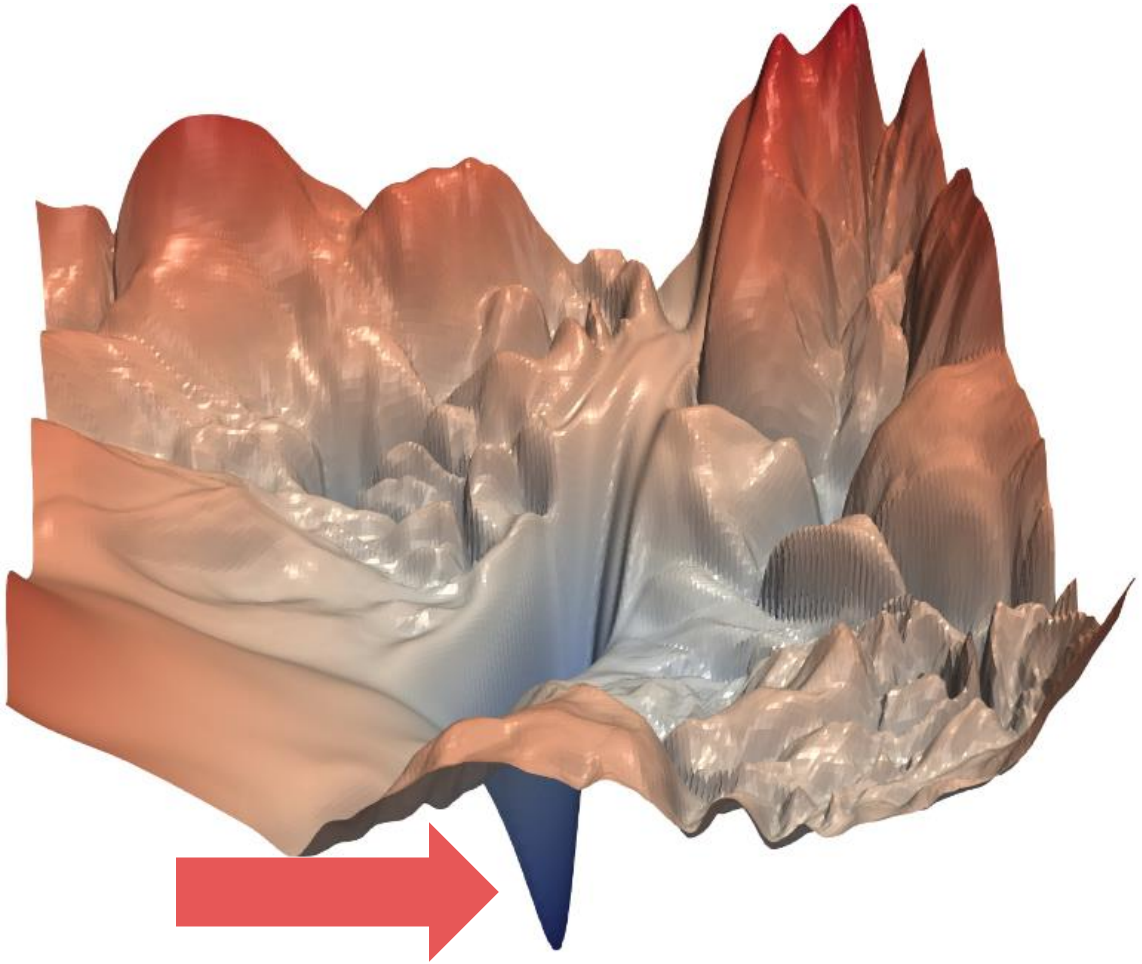
How can we minimize network errors?

- Network error is a function that we want to minimize!



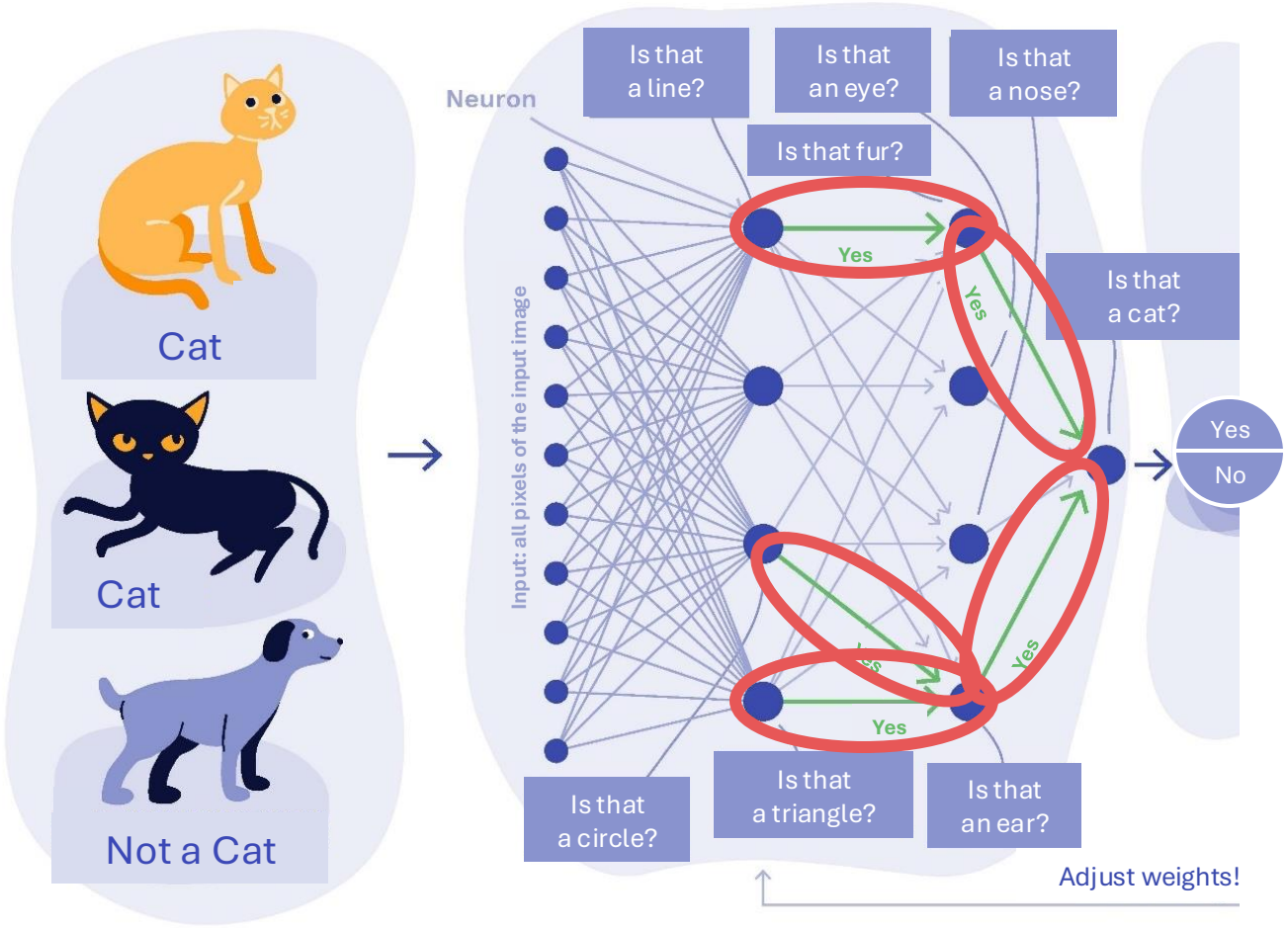
- Find the treasure at the deepest point!

More realistic visualization of the error landscape of neural networks



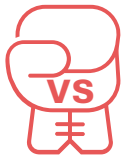
Li et al. (2018). Advances in neural information processing systems. 31.

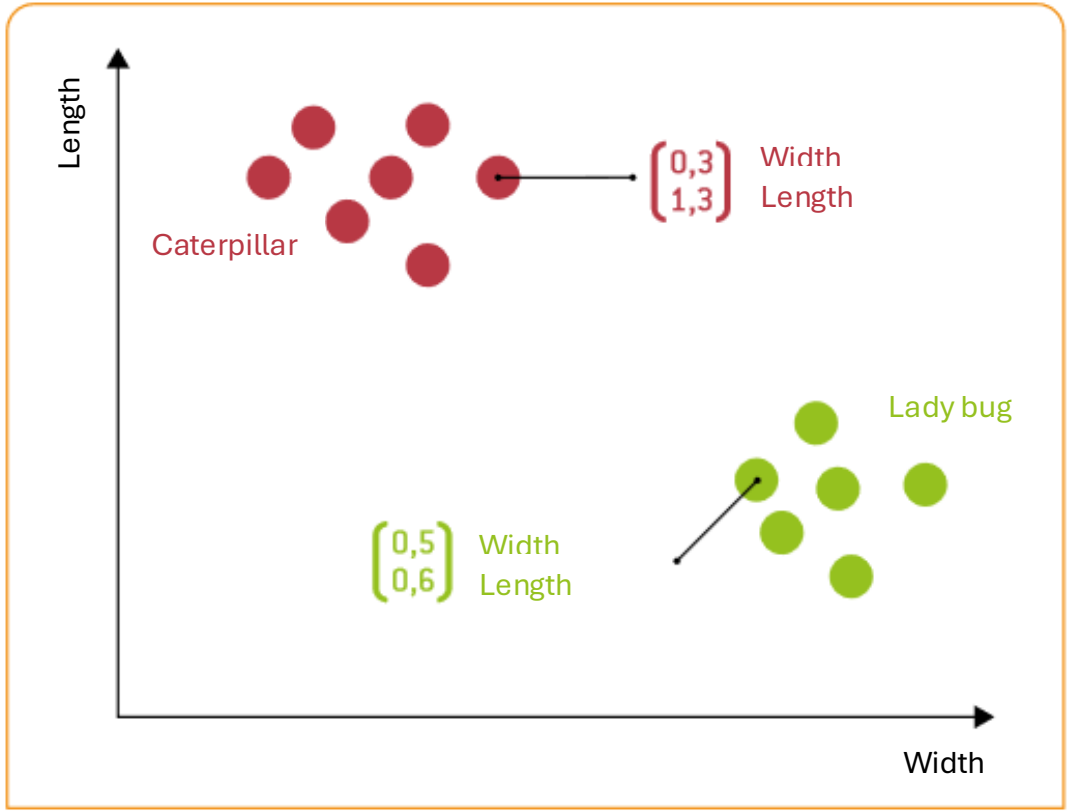
But we still need something else…

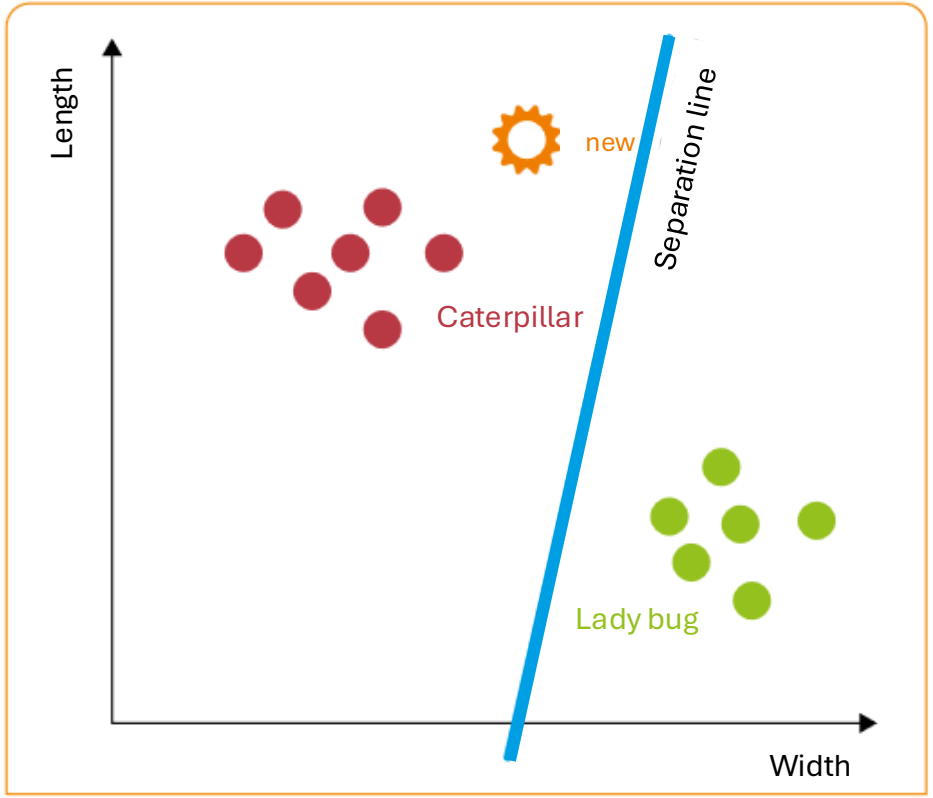


- … namely the so-called “activations”
- Let's take a look at how we calculate these.

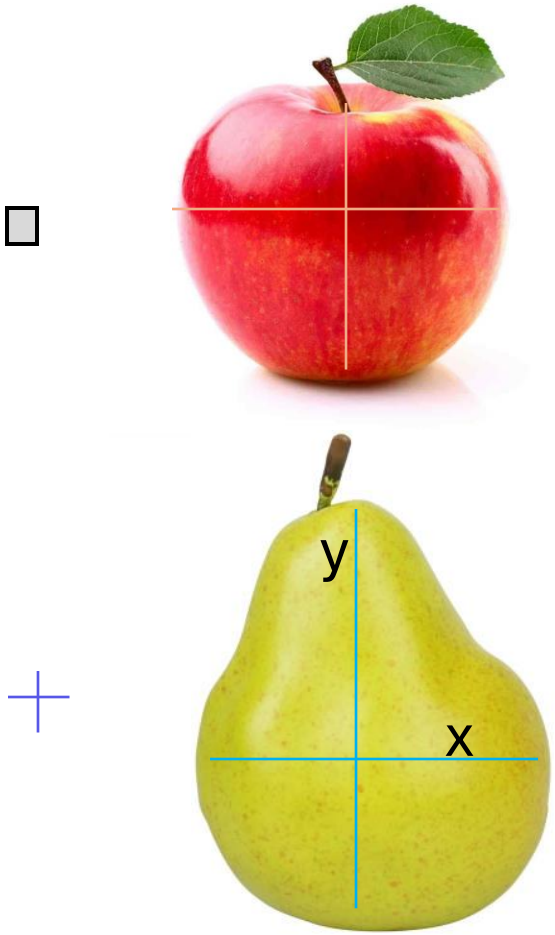
But why do we need these “activations” – An example

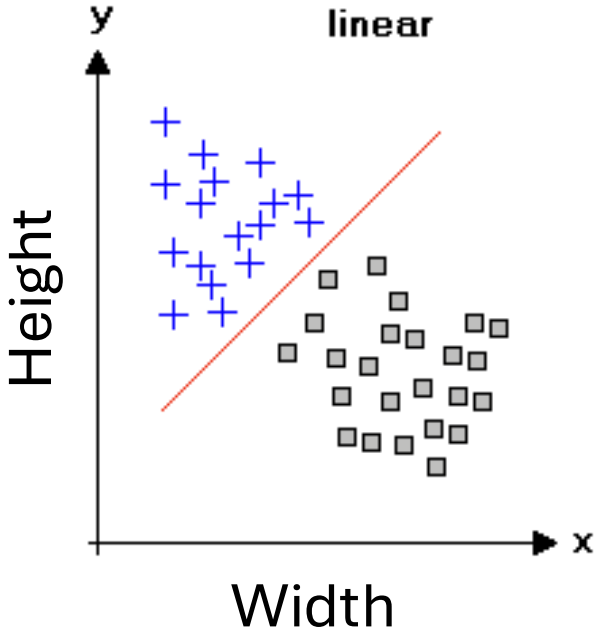
Lady bug  **Caterpillar**





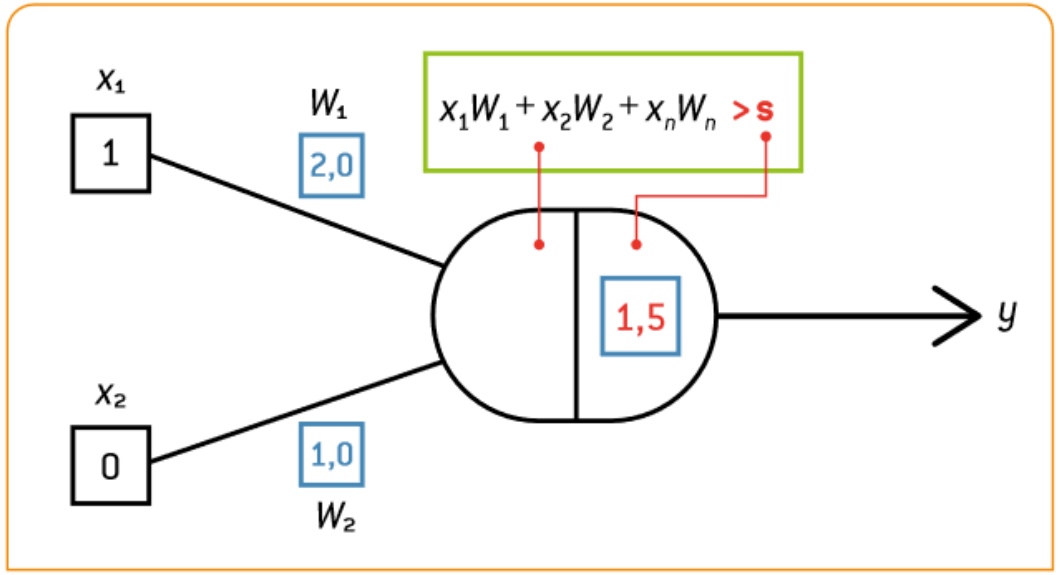
But what happens if we cannot separate the samples linearly?



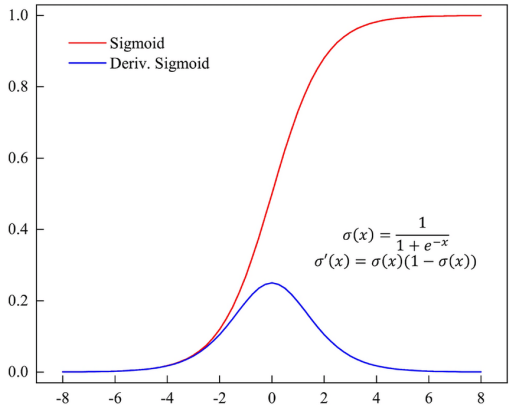


https://studymachinelearning.com/activation-functions-in-neural-network/

Nonlinear activation functions enable us to learn complex patterns

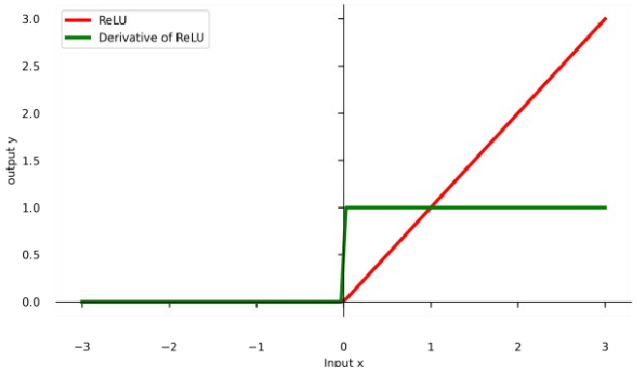


Continuous functions and their derivatives:



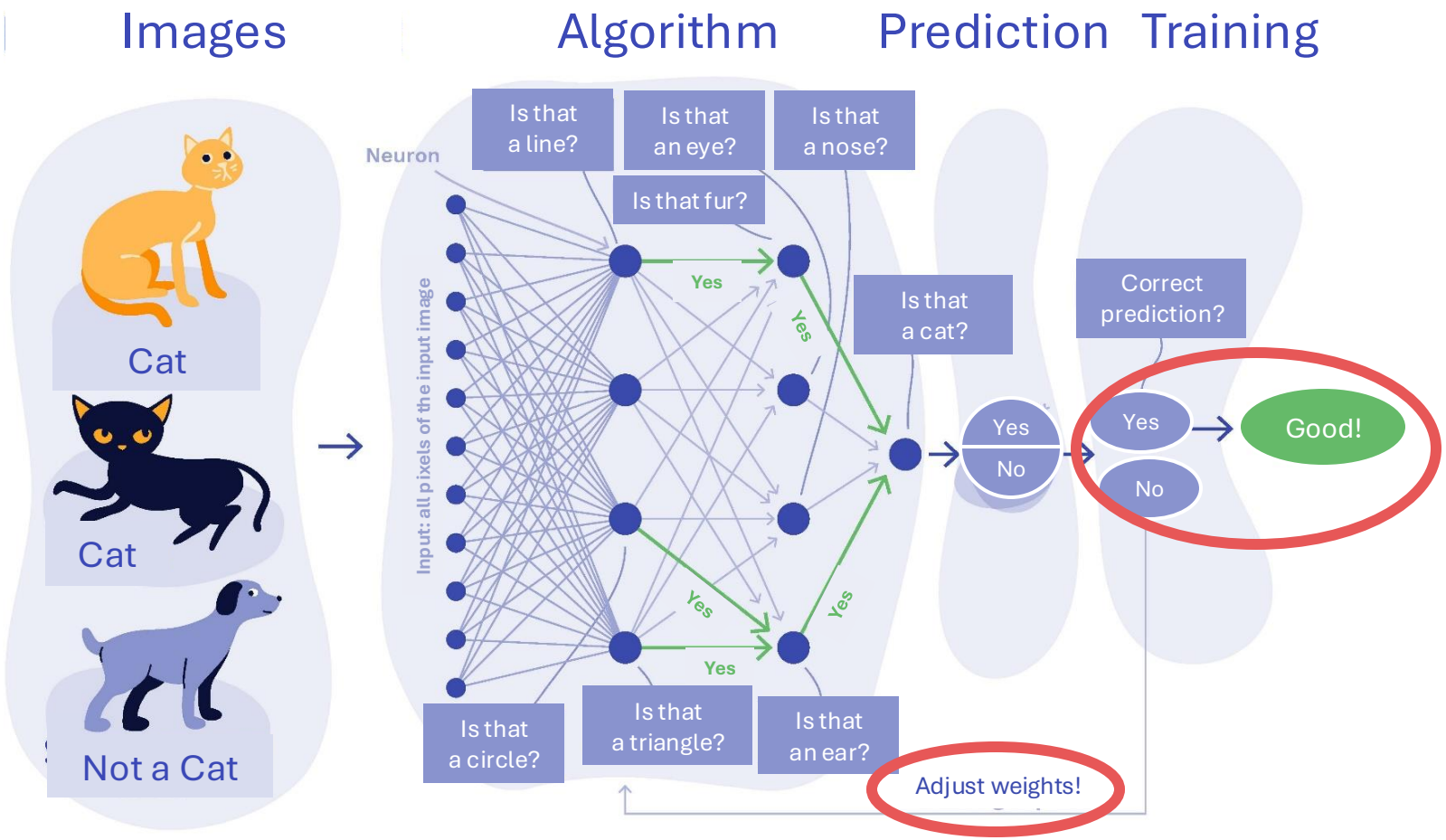
Sigmoid

ReLU

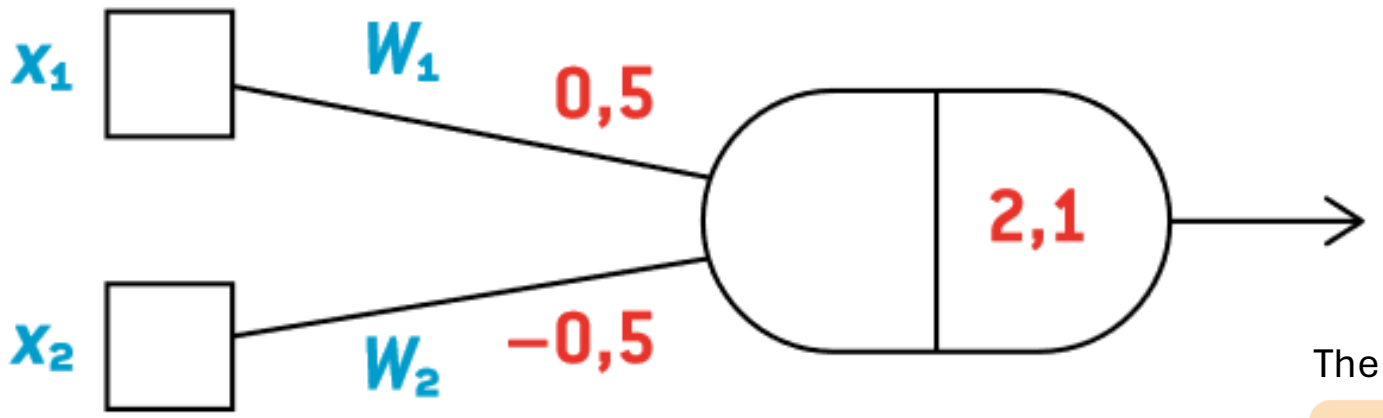


High-performance neural networks do not use a binary threshold function.

Now let's take another look at the learning rule



Learning rule



Actual output: o

Desired output: y

The learning rule (simplified here) is:

$$W_{new} = W_{old} + \eta * (y - o) * x_n$$

- $x_1..x_n$: Inputs
- y: Desired output
- o: Actual output
- η: Learning rate, here a fixed value for simplicity's sake, namely 0.5
- $W_1..W_n$: Weight that will be adjusted

Take-Home

Key trends

- Deep learning has become increasingly useful over time
- We have growing amounts of training data
- We have increasing computing capacity
- We have increasingly available (freely accessible) software frameworks
- Deep learning has tackled increasingly complex applications with increasing accuracy

Current challenges

- Robustness
- Generalizability
- Explainability
- Un-/Selfsupervised Learning
- Data efficiency
- Ethical and legal issues
- No clear path to “conscious”/strong AI

Potential risks & benefits

- Surveillance and persuasion
  - Freedom of speech, privacy, …
- Biased decision-making
  - Social media, elections, law enforcement
  - Potential discrimination based on ethnicity, gender, sexual orientation, …
- Impact on work culture
  - Increased productivity, change in the profile of the human workforce
- Safety-critical applications
  - Self-driving cars, health care, …
- Cyber security
  - Personalized phishing attacks, blackmail, …
- Lethal autonomous weapons

Questions